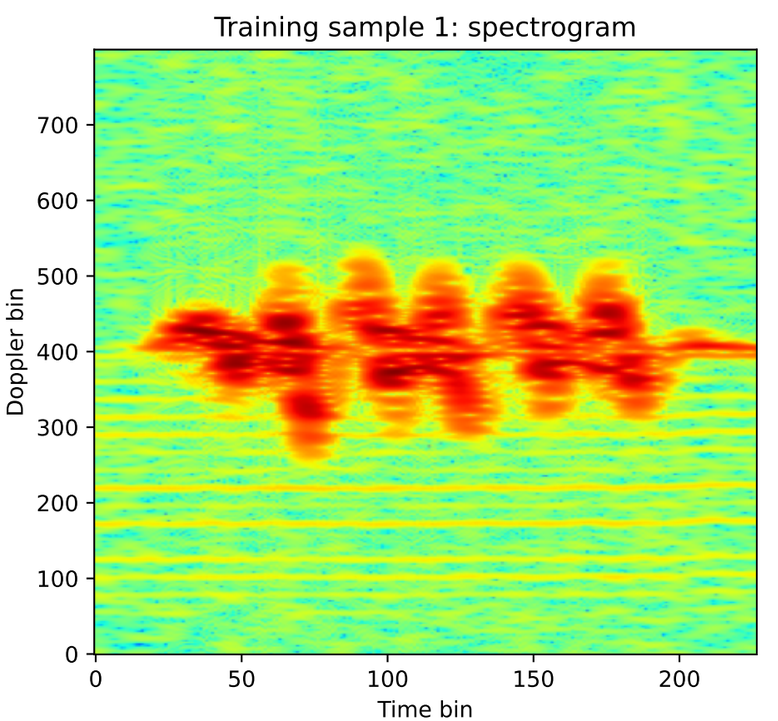
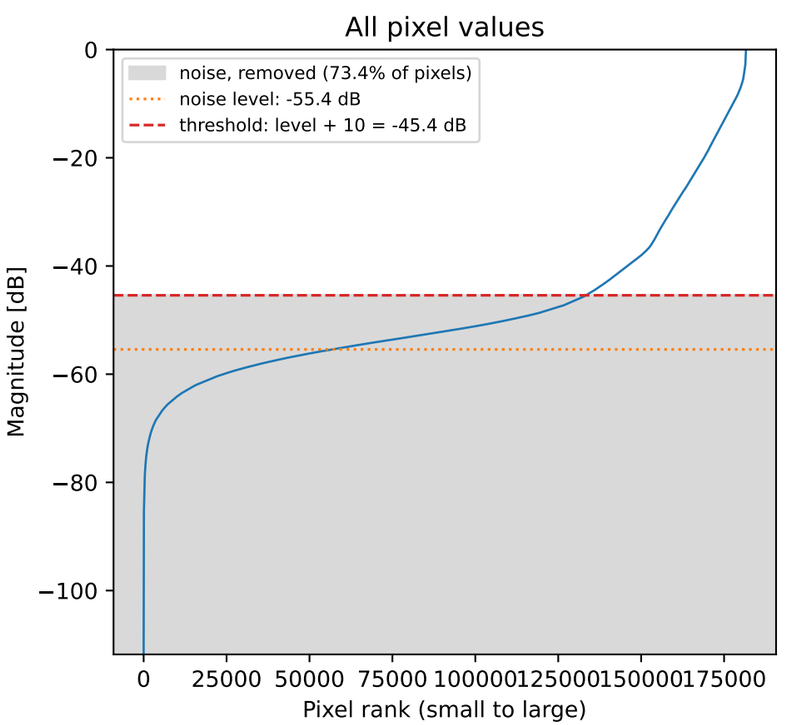
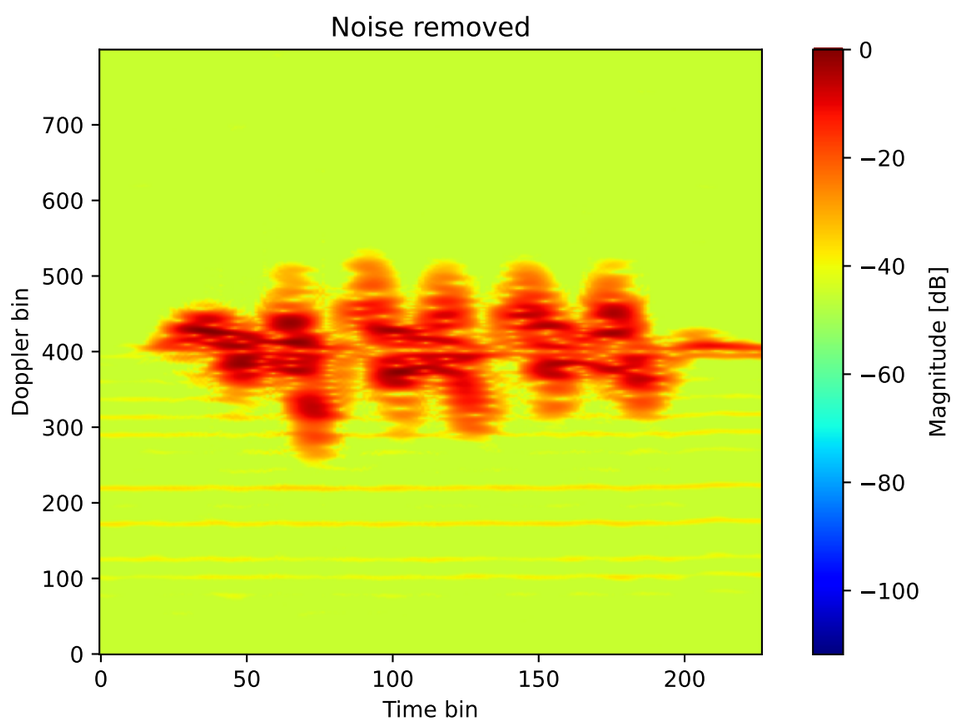
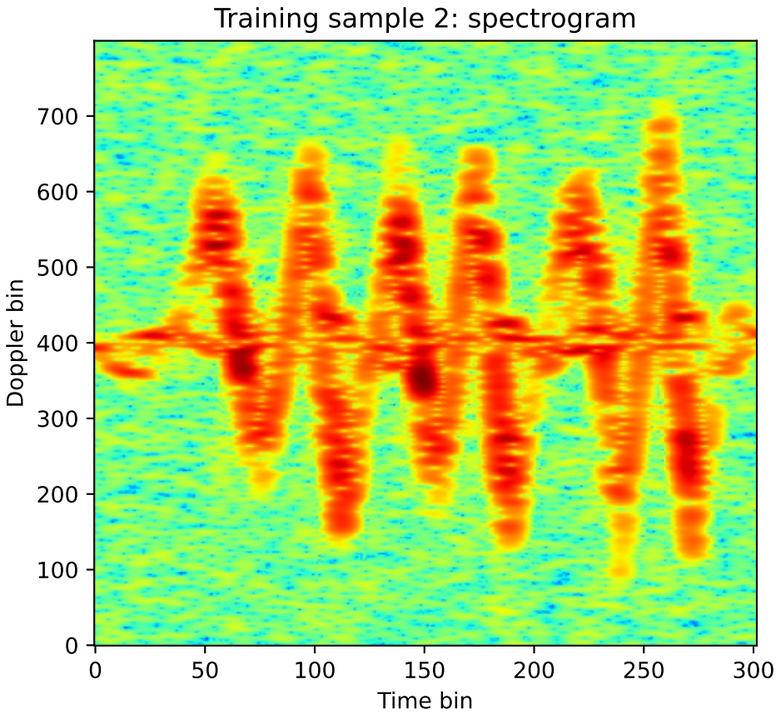
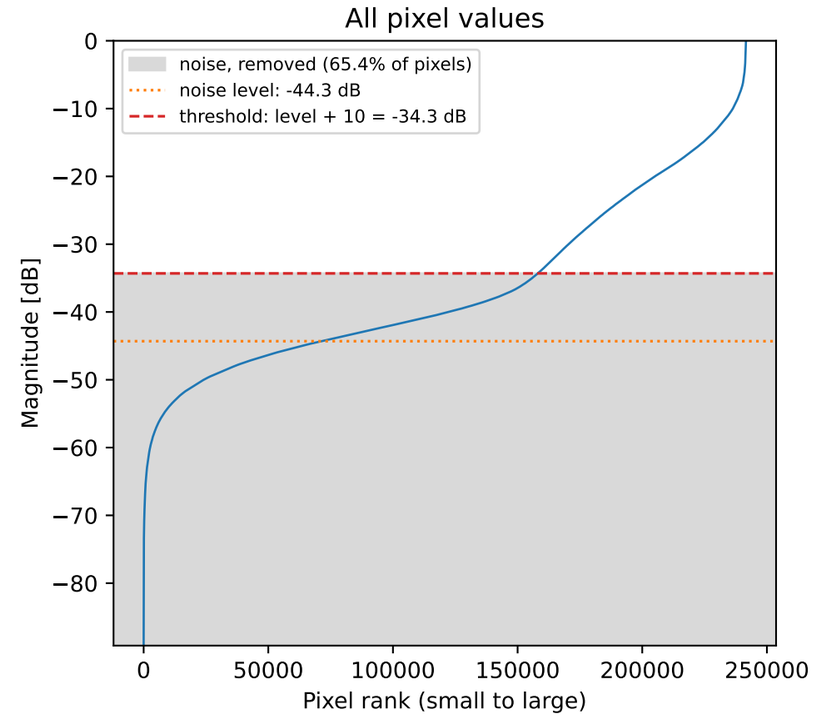
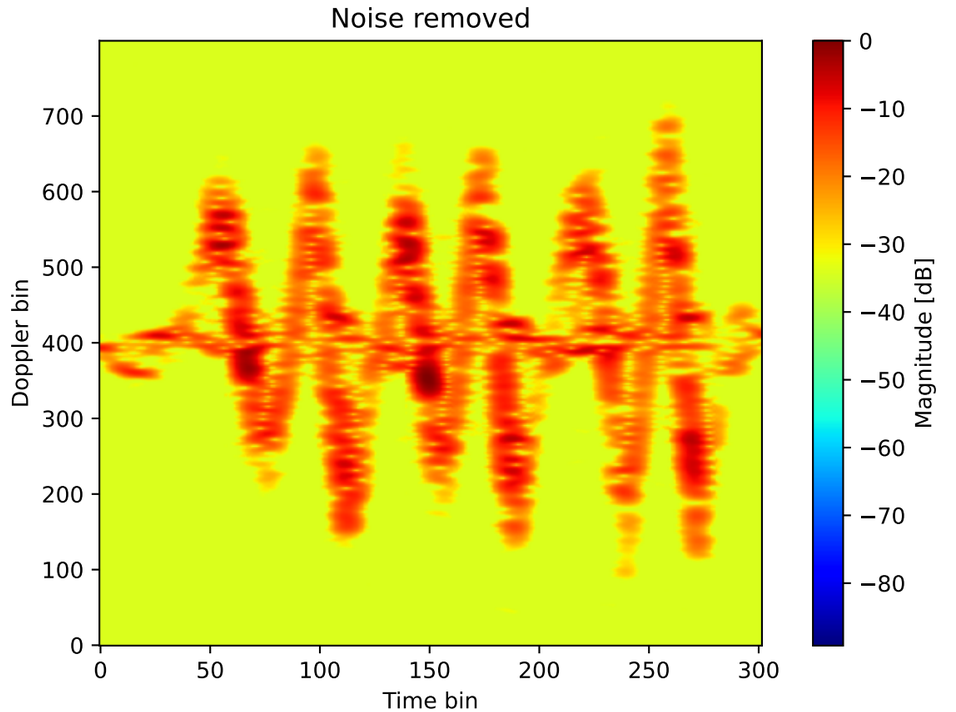

Training sample 1: spectrogram
Doppler bin
Time bin
0
100
200
300
400
500
600
700
0
50
100
150
200


All pixel values
noise, removed (73.4% of pixels)
noise level: -55.4 dB
threshold: level + 10 = -45.4 dB
Magnitude [dB]
Pixel rank (small to large)
0
−20
−40
−60
−80
−100
0
25000
50000
75000
100000
125000
150000
175000


Noise removed
Doppler bin
Time bin
Magnitude [dB]
0
100
200
300
400
500
600
700
0
50
100
150
200
0
−20
−40
−60
−80
−100


Training sample 2: spectrogram
Doppler bin
Time bin
0
100
200
300
400
500
600
700
0
50
100
150
200
250
300


All pixel values
noise, removed (65.4% of pixels)
noise level: -44.3 dB
threshold: level + 10 = -34.3 dB
Magnitude [dB]
Pixel rank (small to large)
0
−10
−20
−30
−40
−50
−60
−70
−80
0
50000
100000
150000
200000
250000


Noise removed
Doppler bin
Time bin
Magnitude [dB]
0
100
200
300
400
500
600
700
0
50
100
150
200
250
300
0
−10
−20
−30
−40
−50
−60
−70
−80